Sakshi Khatri

FPGA Design Engineer

RESUME_1440

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Total Experience: 8 years

PROFESSIONAL SUMMARY

FPGA Design Engineer with 8 years of experience in fpga, using VHDL, Verilog, Xilinx Vivado, Intel Quartus. Experienced in rtl simulation, fixed-point arithmetic and measurable delivery outcomes.

TECHNICAL SKILLS

Skills: AXI4, Intel Quartus, Timing Closure, VHDL, Xilinx Vivado, Verilog, DSP, SPI, Python, Hardware Simulation, FPGA, UART

Methods: RTL simulation, Fixed-point arithmetic, Board bring-up

PROFESSIONAL EXPERIENCE

Senior FPGA Design Engineer | ZenithTrail Technologies | Jan 2023 - Present

- Verified 355 protocol corner cases with repeatable simulation suites
- Improved maximum operating frequency by 16% after timing-path optimization
- Lowered end-to-end data-path latency by 36% using pipelined logic
- Coordinated engineering design reviews, acceptance criteria and test evidence with cross-functional stakeholders.

FPGA Design Engineer | Meridian Vector Labs | Jan 2021 - Dec 2022

- Lowered end-to-end data-path latency by 36% using pipelined logic
- Reduced FPGA resource utilization by 21% through arithmetic sharing

Associate FPGA Design Engineer | WillowGrid Labs | Jul 2018 - Dec 2020

- Verified 755 protocol corner cases with repeatable simulation suites
- Improved maximum operating frequency by 38% after timing-path optimization

PROJECTS

Real-Time Signal Filter | AXI4, VHDL, DSP, FPGA

- Implemented fixed-point FPGA pipelines with testbench golden models.
- Documented acceptance criteria and evaluated the design using fixed-point arithmetic and board bring-up.

Protocol Adapter Core | Timing Closure, Hardware Simulation, DSP, VHDL

- Developed reusable SPI and UART modules with loopback tests.
- Documented acceptance criteria and evaluated the design using fixed-point arithmetic and board bring-up.

EDUCATION

B.Tech in Electronics and Communication Engineering, Silverlake School of Engineering, 2018

Relevant Methods: RTL simulation, Fixed-point arithmetic, Board bring-up

CERTIFICATIONS

Digital Signal Processing Basics (2019)